Validity Analysis: MMLU

Target context: South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Overall risk: **HIGH**

Deployment Context

Use case: Task: Environment science and geograpy-specific agricultural knowledge understanding

LLMs evaluated: Generic frontier model vs small llms as well as region specific llms

Domain: Environmental science

Setting: Environment science research utility assessment using llm and local-context

Target population: An environment scientist from Mymensingh, Bangladesh wants to get knowledge about different south asian region specific agricultural practice. So having different region and language specific agricultural and demographic context understanding (e.g. telegu spoken area, hydrabad India and agricultural practice/problems vs farmers practice/problems from Mymensingh, Bangladesh~bengali spoken in Mymensingh accented dialect.)

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	1	Serious concern	high
Input Content ⚠	1	Serious concern	high
Input Form ⚠	1	Serious concern	high
Output Ontology ⚠	1	Serious concern	high
Output Content ⚠	1	Serious concern	high
Output Form ✓	2	Significant gaps	medium
Average	**1.2**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MMLU is fundamentally misaligned with the South Asian Agricultural and Environmental Science deployment across five of six dimensions. Five HIGH-priority dimensions (Input Ontology, Input Content, Input Form, Output Ontology, Output Content) score 1 — the lowest possible — with multiple structural validity violations: complete absence of agro-ecological subjects, US/Anglophone-saturated content, English-only input, MCQ schema unable to encode regional pluralism, and US-anchored ground-truth labels with no South Asian annotator validation. Output Form scores 2, partially supporting cross-model comparison via accuracy metrics. The single biggest concern is that the benchmark cannot, by construction, measure the constructs the user needs to evaluate — there is no item in MMLU's 14,079-question test set addressing haor wetland ecology, boro/aman/aus rice cultivation, char-land farming, Deccan dry-land cropping, or coastal aquaculture. Cross-border Bangladesh–India water-policy questions, which are actively contested in 2025, are entirely absent and could not be represented in the four-option MCQ format even if added. The benchmark is suitable only as a weak generic STEM-reasoning baseline for cross-model comparison.

Practical Guidance

What This Benchmark Measures

MMLU measures English-language MCQ accuracy on US/Western academic and professional knowledge across 57 subjects, with primary signal in mathematical reasoning, formal logic, and physical sciences (the dimensions where it scores least poorly for cultural neutrality). For this deployment, MMLU can provide a baseline cross-model accuracy comparison on culturally-neutral STEM and statistics items (per Strength 1 and Strength 3 in dataset analysis), and calibration diagnostics [Q60-62, Q106] useful for understanding model uncertainty behavior. It cannot measure South Asian agricultural or environmental science knowledge, nor any multilingual capability.

Construct Depth

Construct depth for the deployment's intended construct (South Asian agricultural and environmental science knowledge) is essentially zero: Input Ontology contains no agro-ecological subjects, Input Content contains no regional sourcing, Input Form precludes regional languages, Output Ontology cannot encode regional divergence, and Output Content has no regional annotator validation. Where MMLU does probe deeply (US legal/medical/historical knowledge), the construct is irrelevant to the deployment. The only modest depth is in language-independent STEM reasoning, which provides shallow but stable signal for cross-model comparison.

What Else You Need

Substantial supplementation is required across five dimensions: (1) Input Ontology — construct or adopt a South Asian agricultural knowledge taxonomy (BRRI/BARI/ICAR-aligned); (2) Input Content — recruit BAU, PJTSAU, ANGRAU agronomists to author items in Bengali, Telugu, and English covering haor/delta/dry-land/aquaculture systems; (3) Input Form — build multilingual evaluation infrastructure leveraging IndicTrans2 [WEB-18] and IndicBERT v2 [WEB-25] foundations, with explicit Bangladeshi-Bengali and Mymensingh-dialect coverage; (4) Output Ontology — design schema permitting regional-pluralism annotation for cross-border policy items; (5) Output Content — establish a regional annotator pool spanning Bangladesh and the Indian states, with explicit protocol for nationally-divergent answer keys (informed by current Ganges/Teesta context [WEB-1, WEB-2, WEB-4]). The synthetic agricultural QA methodology in arXiv 2507.16974 [WEB-26] provides a methodological precedent but covers only Hindi/Punjabi.

ID	Evidence
Q60	“We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.” (p.7)
Q106	“While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is r = 0.81, compared to r = 0.63 in the zero-shot setting.” (p.13)
WEB-1	Active Bangladesh–India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)

WEB-18	IndicTrans2 covers 22 Indian languages including Telugu but is a translation system, not knowledge benchmark (https://openreview.net/forum?id=vfT4YuzAYA)
WEB-25	IndicXTREME (ACL 2023) covers Bengali/Telugu for general NLU but no agricultural evaluation (https://aclanthology.org/2023.acl-long.693)
WEB-26	https://arxiv.org/html/2507.16974v2